Smart Healthcare Risk & Disease Analysis

Comprehensive College Capstone Project Report | Data Analytics with Tableau & Python

1. Abstract & Executive Summary

This comprehensive project presents an end-to-end data analytics and visual clinical informatics ecosystem analyzing 5,000 patient records across 16 biometric and lifestyle attributes.

A critical forensic audit revealed that the raw dataset field health_risk_score was degenerate (constant 100.0 across all rows). We resolved this by engineering a validated Calibrated Health Risk Score (0-100) adhering to American Heart Association and American Diabetes Association clinical guidelines.

The project delivers 8 publication-grade charts, an 8-worksheet Tableau workbook (project.twbx) with 2 dashboards and a 4-scene story, a responsive Flask web portal with real-time risk simulator, and 11 passed unit tests.

2. Cohort Demographics & Biomarker Statistics

Metric Name	Observed Value	Clinical Interpretation / Benchmark
Total Cohort Size	5,000 Patients	Sufficient sample size for robust statistical inference
Gender Distribution	50.12% F / 49.88% M	Perfect balance across male and female populations
Age Range	18 to 79 Years (Mean 48.8)	Covers young adults, working age, and senior populations
Mean Systolic Blood Pressure	134.4 mmHg (Std 26.2)	Over 40% of cohort exhibits Stage 1 or Stage 2 Hypertension
Mean Fasting Glucose	134.5 mg/dL (Std 37.4)	Significant prediabetic and diabetic cohort representation
Mean Total Cholesterol	224.6 mg/dL (Std 43.1)	Mean lies above desirable threshold (< 200 mg/dL)
Heart Disease Prevalence	1,499 Cases (29.98%)	Primary cardiovascular outcome endpoint
Diabetes Mellitus Prevalence	1,746 Cases (34.92%)	Primary metabolic outcome endpoint
Stroke Event Prevalence	1,243 Cases (24.86%)	Primary cerebrovascular outcome endpoint
Triad Multi-Morbidity Rate	1,207 Patients (24.14%)	Patients concurrently diagnosed with HD + Diabetes + Stroke

3. Forensic Audit of health_risk_score Anomaly

Automated statistical testing confirmed that raw health_risk_score has a standard deviation of exactly 0.0000, min 100.0, max 100.0, and 1 unique value.

Our engineered Calibrated Health Risk Score restores continuous variance (range: 3.2 to 94.4, mean: 44.55, std: 20.82) and allows legitimate clinical stratification into Low (43.5%), Moderate (26.2%), High (17.3%), and Critical (13.0%) risk tiers.

4. Tableau Business Intelligence Architecture

We created smart_healthcare_analysis.twb and packaged project.twbx featuring:

• 8 Worksheets: KPI Summary, Disease by Age, Risk Tiers, BP vs Glucose, Lifestyle Matrix, Comorbidity Burden, Symptoms, and BMI vs Cholesterol.

• 2 Dashboards: Clinical Risk & Biomarker Dashboard, Population Health & Lifestyle Matrix.

• 1 Analytical Story: 4-scene narrative arc from anomaly audit to targeted care intervention.

5. Software Architecture & Web Implementation

A modern Flask web application (app.py) serves executive metrics, exploratory analysis galleries, local Tableau mirrors, and an interactive real-time Clinical Risk Assessment Calculator.

The frontend is fully responsive across desktop, tablet, and mobile devices using Bootstrap 5, FontAwesome, and Chart.js.

6. Quality Assurance & Testing Suite

An automated pytest suite (tests/) validates raw and cleaned data schemas, constant score detection, Flask REST routes, API endpoints, and Tableau workbook XML structure with 11/11 tests passing.